Windows Event ID - What It Means - When to Use It

15 key Windows Security Event IDs for log analysis

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How to use this sheet. These are 15 key Windows Security Event IDs. To find one, open Event Viewer, go to Windows Logs, then Security, click **Filter Current Log** in the Actions pane, type the Event ID, and click OK. The Logon Type number inside events 4624 and 4625 tells you how the logon happened. Events marked with a lab note were generated on the Windows laptop during Week 2, using a dummy user.

Windows Event ID	What It Means	When to Use It
A. Logon activity		
4624	**An account was successfully logged on.** Windows records this each time an account logs on successfully. It shows the account name and a Logon Type number, for example 2 for a logon at the keyboard and 3 for a network logon.	To confirm who logged in, when, and how. The Logon Type tells you if the logon was local or came over the network. In the Week 2 lab, a correct password for the dummy user over the network gave Logon Type 3.
4625	**An account failed to log on.** Recorded when a logon attempt fails. It shows the account name, the Logon Type, and a Failure Reason with Status codes that explain why it failed.	To find wrong passwords and brute force attempts, many failures in a short time. In the lab, a wrong password for the dummy user produced it with Logon Type 3, and it is the event exported with PowerShell.
4634	**An account was logged off.** Recorded when a logon session ends.	To work out how long a session lasted. Match it with the 4624 event that has the same Logon ID.
4648	**A logon was attempted using explicit credentials.** Recorded when a program logs on with a username and password given for a different account, instead of the account already signed in. Examples are runas, or connecting to a shared folder with other credentials.	To spot one account using another account's credentials, which can be a sign of lateral movement between computers.
4672	**Special privileges were assigned to a new logon.** Recorded when an account with powerful rights, such as an administrator, logs on.	To find which logons had admin-level rights. An unexpected account in this event is a red flag.
B. Process and service activity		
4688	**A new process was created.** Recorded each time a program starts. It shows the program path and the account that started it. Process creation auditing must be switched on in Windows for it to appear.	To track which programs were run and by whom, for example to find a suspicious tool started by an unusual account.
4697	**A service was installed in the system.** The Security log record of a new service being installed.	To catch persistence, where an attacker installs a service to run again after a restart. The related event 7045 in the System log was used in the lab for the dummy service.
C. Account and group changes		

Windows Event ID	What It Means	When to Use It
4720	**A user account was created.** Recorded when a new user account is created. It shows the new account name and the account that created it.	To detect unexpected or backdoor accounts. In the lab, creating the dummy user produced it.
4722	**A user account was enabled.** Recorded when a disabled account is switched back on.	To catch old or unused accounts being re-enabled for misuse.
4724	**An attempt was made to reset an account's password.** Recorded when someone tries to reset the password of an account. This is a reset by another person, not the owner changing their own password.	To check for password resets that the account owner did not ask for.
4732	**A member was added to a security-enabled local group.** Recorded when an account is added to a local group such as Administrators. It shows the member and the group name.	To detect privilege escalation. In the lab, adding the dummy user to Administrators produced it.
D. Kerberos and NTLM authentication		
4768	**A Kerberos authentication ticket (TGT) was requested.** Kerberos is the main logon method in Windows domains. This is logged on the domain controller when a user asks for their first ticket, which happens at logon.	To review domain logons. Many requests for different users in a short time can point to password guessing. It is mostly seen on domain controllers, so it may not appear on a single laptop.
4769	**A Kerberos service ticket was requested.** After the first ticket, a user asks for a ticket to use a specific service. The domain controller logs each request.	To see which services accounts are using. An unusually high number of requests can point to an attack called Kerberoasting.
4776	**The computer attempted to validate the credentials for an account.** Recorded when NTLM, an older logon method, is used to check an account's password. A failed check shows an error code.	To check NTLM logon attempts and failures, for example repeated failures from one source.
E. Log tampering		
1102	**The audit log was cleared.** Recorded in the Security log when someone clears it. It shows the account that cleared it.	Treat it as a red flag, because attackers clear logs to hide their steps. In the lab, clearing the Security log produced it, after a backup copy was saved.